Deal intelligence: from screening to an evidence-backed IC memo

Product case study – personal prototype. Repository: github.com/kakali-bandyopadhyay/private-markets-deal-intelligence · Live dashboard: kakali-bandyopadhyay.github.io/private-markets-deal-intelligence

Synthetic companies, documents and figures. Illustrative screening criteria and policies. Not connected to, or based on, any firm's systems or processes. The system prepares evidence; people make every decision.

1. The problem and the users

Proposition: one workflow that makes evidence, source conflicts, owners and approvals traceable before an investment committee (IC) decides.

Problem hypothesis behind it (not an established industry finding): some of the most expensive late surprises in a deal come from evidence problems rather than missing analysis – a figure in the IC memo nobody can trace, two adviser reports that quietly disagree, or a red flag with no owner.

User	Where they lose time or confidence	What they need
Deal team (associate, VP)	Finding which document says what; reconciling management, adviser and analysis figures; re-checking citations before IC	Cited answers, conflicts surfaced early, a memo assembled from sourced claims
Investment committee	Cannot tell a sourced fact from a management claim or an expert-call rumour; sees one number where sources disagree	Every claim traceable; disagreements and the chosen basis visible; open risks owned
Operating / value-creation team	Inherits red flags and assumptions after closing without the evidence behind them	Owned red flags with treatment and source, ready for the 100-day plan

2. The v1 workflow and the choices behind it

One deal (Project Larkspur, a synthetic fleet-maintenance software company) moves through one workflow:

- Screening against configurable criteria: each criterion MET, NOT MET, UNKNOWN or CONFLICTED with its source – no composite score; a conflicted criterion is never treated as passed; a person records the decision to advance.
- Evidence changes: the restated memorandum (revenue EUR 92m to 88m) and later adviser reports, shown version by version.
- Diligence Q&A: every claim cites a passage; disagreements are shown, never resolved by the system; questions about other companies are refused with a reason.
- Conflicts: three live disagreements (recurring revenue 80% vs 71%, adjusted EBITDA EUR 24m vs 21.5m, and net revenue retention (NRR) – revenue this year from last year's customers as a share of their revenue last year – 112% vs 104%). A named person accepts a base case with a rationale; the conflict stays visible.
- IC memo gate – six process controls on the memo, separate from the screening criteria: G1 every claim cites a valid passage; G2 no citation to a replaced document version; G3 claims on conflicted facts have a recorded decision; G4 every red flag has an owner and a due date; G5 the reviewer is not the drafter; G6 claims resting only on expert calls are labelled unverified leads.
- Human IC decision, recorded against the exact memo version; any later edit makes it stale.

Passing the gate means the memo's evidence is in order, not that the deal meets the investment criteria. Larkspur's memo passes all six controls, yet on the accepted base case NRR is 104%, still below the illustrative 105% screening criterion – the memo says so, and the IC weighs it.

The IC decision (synthetic, recorded by the IC chair): PROCEED to confirmatory diligence and final bid. Rationale: the thesis is broadly met and the NRR shortfall is priced in rather than ignored. Conditions: (1) a change-of-control waiver from the largest customer before signing; (2) valuation on the adviser's EBITDA of EUR 21.5m and NRR of 104%; (3) a customer call on the possible 2027 re-tender before the final bid. The screening threshold informs the IC's judgement; it does not decide the outcome mechanically.

Deliberately left out of v1, and why:

- Sourcing feeds, CRM and relationship mapping – they need live data and integrations a prototype cannot show honestly, and trust in the evidence trail comes first.
- Automated fact extraction – facts are hand-tagged so the workflow can be judged on its own; extraction with human confirmation is the first "next" item.
- Embeddings – keyword search scoped to the deal is explainable and reproducible; the held-out misses (section 4) are the trigger to test them.
- Model-first answers – the deterministic mode is the default; the optional model mode only phrases answers from retrieved passages, with citation validation and a cost cap.

Two contrast cases. Project Bramble stops at screening: four of six criteria are UNKNOWN, above the limit of two. Project Osier is blocked at the memo gate for four reasons: G1 – one claim ("market leader in the Nordics") has no citation; G3 – the NRR claim relies on an unresolved conflict (108% management vs about 100% expert); G4 – the tax-inquiry red flag has no owner; G5 – the same person drafted and reviewed the memo.

3. Roadmap and how competing requests are decided

Phase	Scope	User problem
Now (v1)	Screening with evidence, cited Q&A, conflicts and versions, memo gate, recorded IC decision	"I can't trace this number" / "we missed that two reports disagree"
Next	Address the observed misses and evaluate the fixes on a new, unseen set; gated memo export; extraction with human confirmation; access control; red flags into the 100-day plan	Manual tagging; IC-pack preparation; hand-off to the operating team
Later	Sourcing signals and CRM; licensed comparables; portfolio monitoring against the IC case	Finding targets; checking the investment case after closing

Priority order when needs collide: (1) IC evidence standards – citations, conflicts, owners – are not traded away; (2) deal-team speed in diligence; (3) operating-team requests are sequenced after the core workflow proves adoption.

Worked example (a prioritisation decision, not a built feature). The deal team asks for one-click export of the memo to the IC deck, including before the gate passes; the operating team asks for red flags to feed a 100-day plan tracker. Decision: export is sequenced first in Next, scoped to gate-passed memos with the fingerprint on every page – it serves the primary user without weakening the evidence standard. The 100-day feed follows it, because it adds a new user group before the core workflow has proved adoption.

4. Evaluation and value

How each question set was used: the development set (30) was written alongside the rules; held-out run 1 (10 questions, AI-drafted and reviewed by me before the run; the rules had not been tuned to them) was run once, and its misses informed the fixes; run 2 (8 new questions, prepared by an AI assistant that had not inspected the implementation) was run once on the frozen build after the fixes; run 3 (10 new questions, drafted from the source documents; I reviewed the expected labels against those documents before the run, and the questions were not used to tune the implementation before it) was run once on the build fixed after run 2. Each score shows something different, and they are not combined.

Development and held-out results are reported separately, per run and per question. All sets are small: prototype evaluation, not investment-grade validation.

Measures (denominator in brackets): correct outcome – answer type matches the label [all questions]; citation resolves – the cited passage exists, belongs to the deal and predates the as-of date, saying nothing about support [all citations]; citation supports – cited passages include every labelled supporting passage and nothing outside the labelled set [questions to answer]; declined correctly / wrongly answered – refusals for the labelled reason, or answers given instead [questions to refuse]; conflicts shown – every disagreeing value shown [questions labelled conflicted].

Measure	Development (30)	Run 1 (10) at 2f722ee*	Run 2 (8) at 197d4b1	Run 3 (10) at 96a201b**
Correct outcome	30/30	4/10	6/8	7/10
Citation resolves	41/41	12/12	12/12	11/11
Citation supports	19/19	2/6	4/6	2/6
Declined correctly	11/11	2/4	1/2	3/4
Wrongly answered	0/11	2/4	1/2	1/4
Conflicts shown	6/6	0/2	1/2	2/2

The first held-out result, 4/10, exposed the gap: the development score measured rules against questions written alongside them. Every one of the 12 citations resolved to a real, in-scope passage, but only 2 of the 6 answered questions cited the passages that support the answer. Six questions were missed, from five causes:

- H08 – "How much revenue is contracted to renew next year?" was answered with total revenue: the metric matched and the qualifier was ignored (answered a different question).
- H10 – "What share of customers would accept a 6% increase?" was answered from one customer interview (over-claim from one expert call).
- H03, H04 – retention and recurring-revenue questions went to passage search, which skipped the conflict check (one cause, two misses).
- H06 – "updated CIM" was not recognised as a version question (change vocabulary).
- H02 – "levers for growing the business" did not match "plans" and "growth" (paraphrase).

** Code 9cb20c6 plus the ten questions only. ** Frozen at beab828; 96a201b differs only in documentation and an output label.*

After run 1, each cause was fixed and tested with new phrasings. In development checks the run-1 questions now score 10/10, but they were used to find the causes, so run 1 stays published as 4/10.

Run 2 scored 6/8 on the frozen build, with eight new questions I had not seen. All 12 citations resolved to real, in-scope passages; three questions showed issues:

- H2A-08 – false conflict: "year-end net working capital" should have been declined, but generic words ("year", "end", "net") matched a retention passage and its conflict was attached to an unrelated question.
- H2A-04 – missed conflict: a retention question with extra wording went to passage search, fell short of the match bar and was declined instead of falling back to the retention figures and their conflict.
- H2A-05 – right outcome, unsupported citation: the version change was correct, but the cited "explanation" passage was about revenue, not the recurring-revenue share.

Each cause was then fixed; checks after the fixes are development checks, not held-out scores, and run 2 is not re-scored.

Run 3 scored 7/10, with all 11 citations resolving and both conflicts shown. Four questions were missed:

- H3-02 – right outcome, wrong answer: asked what the biggest customer can do on a change of ownership, it returned that customer's 9% revenue share. The fallback added after run 2 caused it – a fix creating a new failure, which is why every run is reported separately.
- H3-01 – paraphrase: "scale of work and spend to bring the code up to date" did not match "remediate", "18 to 24 months", "EUR 3m to 4m".
- H3-05 – vocabulary: "sales" was not recognised as revenue, so the version history was not consulted.
- H3-07 – over-answer: a hypothetical recalculation (excluding support revenue too) returned the existing 80% vs 71% conflict instead of a refusal.

Across three held-out runs the system never cited a passage that did not exist or belonged to another company; its misses are routing and vocabulary, and the most dangerous kind – a confident answer to a different question – appeared in runs 1 and 3. The run-3 case has since been addressed with a targeted fix, checked only in development tests; it still needs validation on a new held-out set before any wider use.

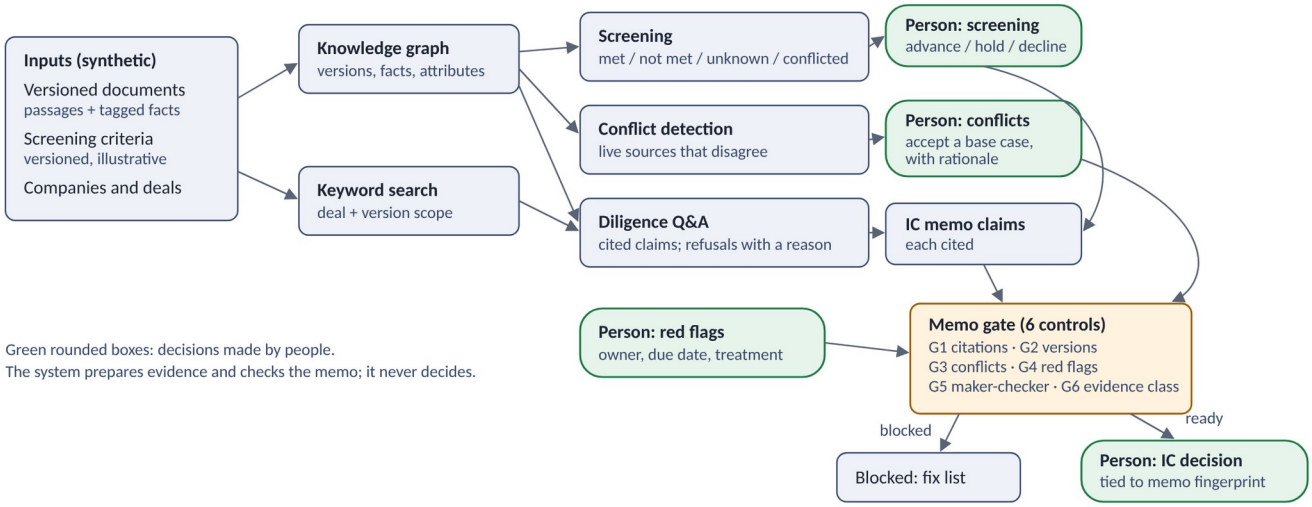
Model mode (an optional model, gpt-5.4-mini, phrasing answers only from retrieved passages) was evaluated on the development set only: 28/30 outcomes at USD 0.023 for 25 calls, with every citation resolving and every refusal correct – but it flattened two conflicts by picking one figure, against its instructions. The conflict check is now deterministic and cannot be overridden by the model; the development re-run scored 29/30 with 6/6 conflicts shown. The no-key deterministic mode stays the default: the model adds readable wording, not accuracy. All held-out results above are from the deterministic mode.

Value, hours first. Illustrative assumptions, to be replaced by timed pilot measurements:

- Assumed, not measured: hours saved per deal 7.3 (low) / 37.0 (base) / 90.5 (high); at an assumed 12 deals a year in diligence, 88 / 444 / 1,086 hours a year.
- Base-case assumptions: evidence look-ups 150 × (12 – 4) min = 20.0 h; conflict reconciliation 6 × (180 – 90) min = 9.0 h; memo assembly and citation checks (960 – 480) min = 8.0 h.
- The range is wide because time per look-up is the least certain input – the first thing a pilot should time.

Measured alongside: time to find evidence; citation resolution and support; conflicts unresolved at IC; correction effort; weekly active users; share of IC memos built through the gate.

5. Architecture (data flow) and limitations



Limitations and what would need validation before real use:

- Facts are hand-tagged; real use needs extraction with human confirmation and a measured error rate.
- Keyword retrieval with a small hand-written synonym list; paraphrase in general is untested beyond the held-out runs.
- Single currency and fiscal year; synthetic roles stand in for people, access control and information barriers.
- Screening criteria, conflict rules and gate controls would be set and approved by the investment team.
- Model mode is evaluated only on the development set and needs model-risk, retention and provider-terms review.

A Codex-assisted repository review against a definition-of-done list identified a memo-gate bypass and an outdated dashboard; both were fixed and re-checked, and a clean clone passed all 87 tests. I reviewed the case study, and run 3's questions and labels, against the source documents.

6. 90-second demo

The live dashboard shows this journey step by step, the gate controls and the full evaluation-run table.

- 0–25 s: the labels (synthetic data, illustrative criteria, people decide); screening on 12 May – four criteria met with sources, two unknown, recorded decision to advance.
- 25–45 s: "What is net revenue retention?" – 112% vs 104%, both cited; the conflict stays visible, and the deal team records 104% as the working case with a rationale; 104% is below the 105% criterion.
- 45–65 s: Osier blocked by four named controls; Larkspur passes all six memo controls with the accepted conflicts still visible.
- 65–85 s: the IC chair records PROCEED with three conditions and a rationale – the NRR shortfall is priced in – tied to the memo fingerprint.
- 85–90 s: 30/30 on the development set; 4/10, 6/8 and 7/10 on three held-out runs – shown separately, with every miss and its cause.